Recordall®
RCDL | Mechanical Flow Meters

# Recordall® Disc Meters

## Lead-Free Bronze Alloy 120 & 170, Sizes 1-1/2 in. (40 mm) & 2 in. (50 mm), NSF/ANSI/CAN Standards 61 and 372 Certified

## DESCRIPTION

Recordall Models 120 and 170 Disc Series meters meet or exceed the most recent revision of AWWA Standard C700 and are available in a lead-free bronze alloy. Both meters comply with the lead-free provisions of the Safe Drinking Water Act, are certified to NSF/ANSI/CAN Standards 61 and 372 (Trade Designations: M120-LL and M170LL) and carry the NSF-61 mark on the housing. All components of the lead-free bronze alloy meter (housing, measuring element, seals, and so on) comprise the certified system.

**Applications:** For use in measurement of potable cold water in residential, commercial and industrial services where flow is in one direction only.

**Operation:** Water flows through the meter's strainer and into the measuring chamber where it causes the disc to nutate. The disc, which moves freely, nutates on its own ball, guided by a thrust roller. A drive magnet transmits the motion of the disc to a follower magnet located within the permanently sealed register. The follower magnet is connected to the register gear train. The gear train reduces the disc nutations into volume totalization units displayed on the register or encoder face.

**Operating Performance:** The Recordall Disc Series meters meet or exceed registration accuracy for the low flow rates (95%), normal operating flow rates (100 ± 1.5%), and maximum continuous operation flow rates as specifically stated in AWWA Standard C700.

**Construction:** Recordall Disc meter construction, which complies with ANSI/AWWA standard C700, consists of three basic components: meter housing, measuring chamber, and permanently sealed register or encoder. The water meter is available in a lead-free bronze alloy. A corrosion-resistant engineered polymer material is used for the measuring chamber.

**Magnetic Drive:** Direct magnetic drive, through the use of high-strength magnets, provides positive, reliable and dependable register coupling for straight-reading or AMR/AMI meter reading options.

**Tamper-Proof Features:** Unauthorized removal of the register or encoder is inhibited by the option of a tamper detection seal wire screw, TORX® tamper-resistant seal screw or the proprietary tamper-resistant keyed seal screw. Each can be installed at the meter site or at the factory.

**Maintenance:** Badger Meter Recordall Disc Series meters are designed and manufactured to provide long-term service with minimal maintenance. When maintenance is required, it can be performed easily either at the meter installation or at any other convenient location.

To simplify maintenance, the register, measuring chamber, and strainer can be replaced without removing the meter housing from the installation. No change gears are required for accuracy calibration. Interchangeability of parts among like-sized meters minimizes spare parts inventory investment. The built-in strainer has an effective straining area of twice the inlet size.

**Connections:** Companion flanges in cast iron or NL bronze are available as options. Straight connection sets are available in NL bronze.



M120 M170

## SPECIFICATIONS

| Meter Model | M120 | M170 |
|---|---|---|
| **Typical Operating Range (100% ± 1.5%)** | 2.5–120 gpm (0.57–27 $m^3$/hr) | 2.5–170 gpm (0.57–39 $m^3$/hr) |
| **Low Flow (Min. 95%)** | 1.25 gpm (0.28 $m^3$/hr) | 1.5 gpm (0.34 $m^3$/hr) |
| **Maximum Continuous Operation** | 80 gpm (18 $m^3$/hr) | 100 gpm (23 $m^3$/hr) |
| **Pressure Loss at Maximum Continuous Operation** | 4.8 psi at 80 gpm (0.33 bar at 18 $m^3$/hr) | 3.3 psi at 100 gpm (0.23 bar at 23 $m^3$/hr) |
| **Maximum Operating Temperature** | 80 °F (26 °C) | 80 °F (26 °C) |
| **Maximum Operating Pressure** | 150 psi (10 bar) | 150 psi (10 bar) |
| **Measuring Element** | Nutating disc, positive displacement | Nutating disc, positive displacement |
| **Meter Connections** | 1-1/2 in. AWWA two-bolt elliptical flange, drilled or 1-1/2–11-1/2 NPT internal pipe threads | 2 in. AWWA two-bolt elliptical flange, drilled or 2–11-1/2 NPT internal pipe threads |
| **Test Plugs** | 1 in. NPT test plug (TP) | 1 in. NPT test plug (TP) |

### Materials

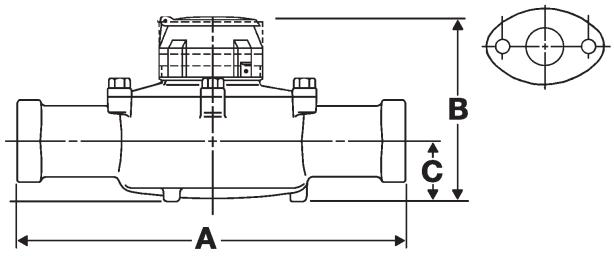
| | |
|---|---|
| **Meter Housing** | Lead-free bronze alloy |
| **Housing Top Plates** | Lead-free bronze alloy |
| **Measuring Chamber** | Engineered polymer |
| **Disc** | Engineered polymer |
| **Trim** | Stainless steel |
| **Strainer** | Engineered polymer |
| **Disc Spindle** | Stainless steel |
| **Magnet** | Ceramic |
| **Magnet Spindle** | Stainless steel |
| **Register Lid and Shroud** | Engineered polymer, bronze |

Badger Meter

RDM-DS-00465-EN-11 (June 2026)

Product Data Sheet

## DIMENSIONS



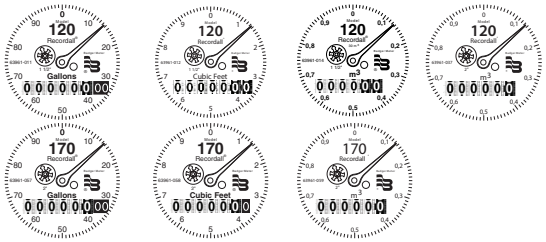

| Meter Size | Meter Model | A Laying Length | B Height Reg./RTR | C Centerline Base | Width | Approx. Shipping Weight |
|---|---|---|---|---|---|---|
| 1-1/2 in. (40 mm) | 120 EL, Hex<br>120 EL, TP | 12-5/8 in. (321 mm) | 7 in. (178 mm) | 2-3/8 in. (60 mm) | 8-3/4 in. (222 mm) | 19 lb (8.6 kg) |
| 1-1/2 in. (40 mm) | 120 ELL<br>120 ELL, TP | 13 in. (330 mm) | 7 in. (178 mm) | 2-3/8 in. (60 mm) | 8-3/4 in. (222 mm) | 19 lb (8.6 kg) |
| 2 in. (50 mm) | 170 EL, Hex<br>170 EL, TP | 15-1/4 in. (387 mm) | 8 in. (203 mm) | 2-7/8 in. (73 mm) | 9-1/2 in. (241 mm) | 30 lb (13.6 kg) |
| 2 in. (50 mm) | 170 ELL<br>170 ELL, TP | 17 in. (432 mm) | 8 in. (203 mm) | 2-7/8 in. (73 mm) | 9-1/2 in. (241 mm) | 30 lb (13.6 kg) |

EL = Elliptical  ELL = Elliptical Long  Hex = Hexagon, 1-1/2–11-1/2 in. NPT Thread  TP=Test Plug 1 in.

## REGISTERS / ENCODERS

### Standard—Sweep-Hand Registration

The standard register is a straight-reading, permanently sealed magnetic drive register. Dirt, moisture, tampering and lens fogging problems are eliminated. The register has a six-odometer wheel totalization display, 360° test circle with center sweep hand, and flow finder to detect leaks. Register gearing is made of self-lubricating engineered polymer, which minimizes friction and provides long life. The multi-position register simplifies meter installation and reading. The register capacity is 100,000,000 gallons (10,000,000 ft³, 1,000,000 m³).




| Meter Model | Gallon | Cubic Feet | Cubic Meter |
|---|---|---|---|
| 120 | 100 | 10 | 1/0.1 |
| 170 | 100 | 10 | 1 |

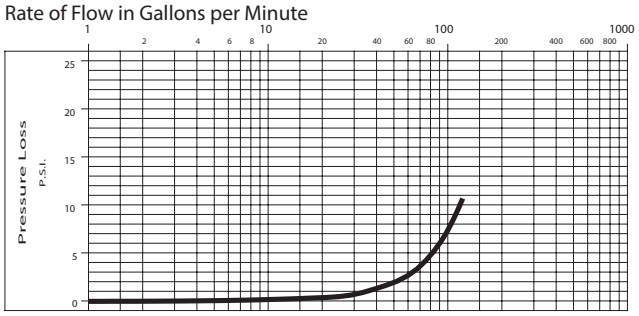
### Optional—Encoders for AMR/AMI Reading Solutions

AMR/AMI solutions are available for all Recordall Disc Series meters. All reading options can be removed from the meter without disrupting water service. Badger Meter encoders provide years of reliable, accurate readings for a variety of applications and are also available pre-wired to Badger Meter approved AMR/AMI solutions. See details at *badgermeter.com*.

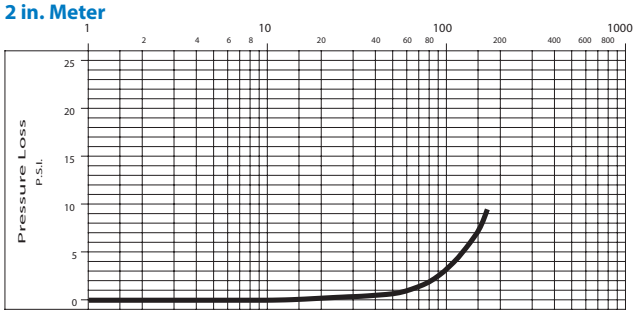
## PRESSURE LOSS CHARTS

### 1-1/2 in. Meter

Rate of Flow in Gallons per Minute



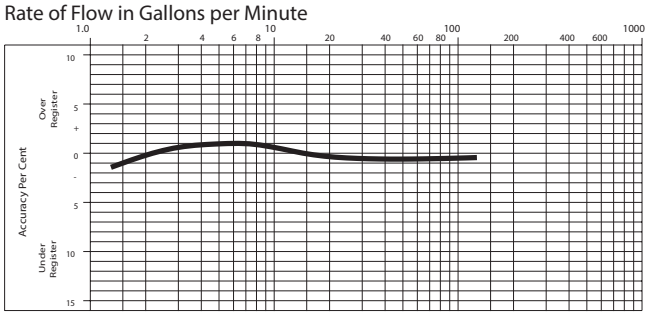

### 2 in. Meter



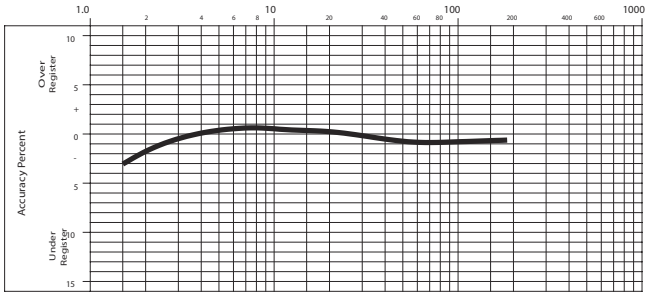

## ACCURACY CHARTS

### 1-1/2 in. Meter

Rate of Flow in Gallons per Minute




### 2 in. Meter




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